

Assignment 2

Assignment Instructions

This assignment covers core concepts in text processing:

✓ **Statement of Purpose (SOP) - and Summary**

✓ **Self-Attention in Transformers**

Submission Guidelines

- **All answers must be handwritten** and submitted using the answer sheet provided at the following [link](#).
- Download and print the answer sheet and write your answers manually on the answer sheet.
- Scan all the pages in the correct sequence, create a PDF document and submit to eClass.
- Please also bring the actual paper submission to the first class after submission and submit it to the professor.
- Ensure your handwriting is **clear and legible**.
- If your handwriting is **not readable**, you will not receive a score for that task.

♦ **Formatting & Presentation**

- **Write neatly and in an organized manner.**
- Clearly label each task and **show all necessary steps**.
- **Illegible or incomplete responses will not be graded.**

Deadline

- **Submission due by: 14.05.2026 Thursday [23:55]**
- **Late submissions:**
 - **within 24 hours: 50% penalty**
 - **Beyond 24 hours: not accepted**

Academic Integrity

- **This is an individual assignment.**
- **Cheating of any form is strictly prohibited.**
- Any form of plagiarism will result in a **zero score** for the assignment
- Not using the corresponding ID-based values at indicated sections will result in 0 score.

Part-1

Fine-Tuning GPT-2 for SOP-Driven Text Generation

The goal of this assignment is to apply concepts learned in class about neural networks and NLP by fine-tuning a pre-trained GPT-2 model on a personal Statement of Purpose (SOP).

The fine-tuned model will then be used to generate text based on seed phrase which is "[Student Name]'s academic interests include", demonstrating understanding and application of machine learning techniques in natural language processing.

You can follow the below steps to perform this assignment.

Write or obtain your SOP:

Prepare a comprehensive SOP for a Master's Degree (**more than 1500 words**). This SOP should detail your academic background, interests, and reasons for pursuing further education.

It is an important document that you can use for your Master's Application.

Upload your SOP to Colab:

Once you have your SOP ready, convert it to a .txt form and upload to google colab. I recommend using colab for this assignment.

Read the SOP from your file:

Implement code to read the SOP text from the uploaded file. Ensure proper encoding is set to avoid errors.

Prepare the dataset:

Use the GPT-2 tokenizer to process your text.

Model Loading and Fine-Tuning:

- Load the pre-trained GPT-2 model.
- Fine-tune this model on your SOP using the prepared dataset.
- Pay attention to setting the correct parameters for the training process such as learning rate and number of epochs.

Implement a text generation function:

After fine-tuning, use the model to generate text based on provided seed phrases.

Apply ROUGE-L evaluation: **show performance result**

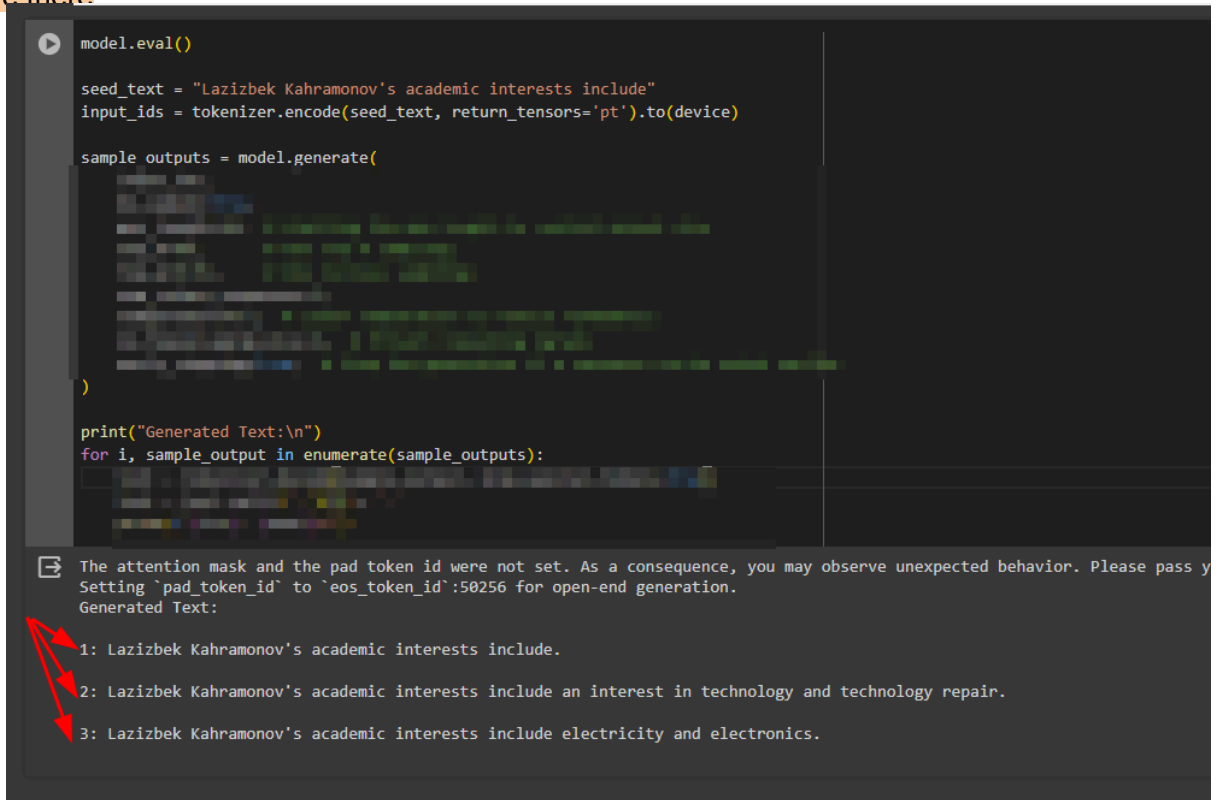
Get 3 results using the seed format provided:

Use "[Student (Full) Name]'s academic interests include" as seed to explore how well the model has adapted to the style and content of your SOP to get 3 generated sentences.

You must use your **own full name**. If you do not use your own name for any reason, you will get **0** for the **entire assignment**. Please find my output on the next page.

You should also have similar output like mine below. It is not perfect and yours does not need to be perfect as well.

When you download your notebook from google colab and submit, **make sure that outputs are there**



```
model.eval()

seed_text = "Lazizbek Kahramonov's academic interests include"
input_ids = tokenizer.encode(seed_text, return_tensors='pt').to(device)

sample_outputs = model.generate(
    input_ids=input_ids,
    max_length=100,
    min_length=10,
    top_k=5,
    num_return_sequences=3,
    do_sample=True,
    eos_token_id=tokenizer.eos_token_id,
)

print("Generated Text:\n")
for i, sample_output in enumerate(sample_outputs):
    print(f"{i}: {tokenizer.decode(sample_output.cpu().numpy())}")
```

The attention mask and the pad token id were not set. As a consequence, you may observe unexpected behavior. Please pass y
Setting `pad\_token\_id` to `eos\_token\_id`:50256 for open-end generation.
Generated Text:

- 1: Lazizbek Kahramonov's academic interests include.
- 2: Lazizbek Kahramonov's academic interests include an interest in technology and technology repair.
- 3: Lazizbek Kahramonov's academic interests include electricity and electronics.

Requirements

All the coding steps must be performed in a jupyter notebook, use colab and download.

Do no cheat!

Do not copy from each other.

Solve the problems individually, not with a friend or a team.

If anyone has the same code, it will be considered cheating and you will get 0 for the entire assignment.

The structure of the notebook.

[Task name] Fine-Tuning GPT-2 for SOP-Driven Text Generation

[Code]

[Result output]

The results should be visible and saved when you submit your jupyter notebook.

What to submit

Submit 2 files:

1) One Jupyter notebook file - **NOT** zipped, named as your student ID.

Example, U1810021.ipynb

2) SOP in the .txt form - **NOT** zipped, named as your student ID.

Example, U1810021.txt or U1810021.doc

3) Also write in

[Not a python file, no separate images, all result outputs should already be shown inside the notebook]

Part 2. Calculate One Pass of Self-Attention

Please watch this awesome [video](#) about attention in transformers and calculate one pass of self-attention step-by-step as follows.

1. Setup: Token Embeddings from Your ID

1. Take the **last 4 digits** of your student ID, call them a,b,c,d.
2. Form your 2-token embedding matrix

$$X = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \quad (\text{so } \mathbf{x}_1 = [a, b], \mathbf{x}_2 = [c, d])$$

2. Fixed Projection Matrices

We'll use small, easy-to-compute weight matrices (all 2×2):

$$W^Q = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad W^K = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad W^V = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

- W^Q “reads” each embedding as a Query.
 - W^K turns embeddings into Keys.
 - W^V turns embeddings into Values.
-

3. Step-by-Step Calculations

Step 1. Compute Q,K,V

$$Q = X W^{Q^\top}, \quad K = X W^{K^\top}, \quad V = X W^{V^\top}.$$

- **Shapes:** $X \in \mathbb{R}^{2 \times 2}$, so $Q, K, V \in \mathbb{R}^{2 \times 2}$.
 - We now have, for each token i , a query vector $\mathbf{q}_i$, a key vector $\mathbf{k}_i$, and a value vector $\mathbf{v}_i$.
-

Step 2. Form the Score Matrix S

$$S = Q K^\top, \quad S_{ij} = \mathbf{q}_i \cdot \mathbf{k}_j.$$

- **Dimensions:** $S \in \mathbb{R}^{2 \times 2}$
 - Each entry S_{ij} measures how “compatible” token i ’s query is with token j ’s key—i.e. how much i should attend to j .
-

Step 3. Scale the Scores

$$\tilde{S} = \frac{S}{\sqrt{d_k}} \quad \text{with } d_k = 2,$$

so numerically divide each element by $\sqrt{2}$.

- Without scaling, large dot-products push the softmax into extremes and hurt gradient flow. Dividing by $\sqrt{d_k}$ keeps the values in a well-behaved range.
-

Step 4. Convert to Attention Weights

$$A = \text{softmax}(\tilde{S}) \quad (\text{apply softmax **row-wise**}).$$

Concretely, for each row i :

$$A_{ij} = \frac{\exp(\tilde{S}_{ij})}{\exp(\tilde{S}_{i1}) + \exp(\tilde{S}_{i2})}.$$

- Softmax turns raw scores into a probability distribution over “which token j each token i attends to.”
-

Step 5. Compute the Output Z

$$Z = AV, \quad Z_i = \sum_{j=1}^2 A_{ij} \mathbf{v}_j,$$

giving $Z \in \mathbb{R}^{2 \times 2}$

- Each output row Z_i is a **contextualized embedding** for token i : a weighted blend of all value vectors.
-

4. Understand the final results

- The final matrix Z is exactly your **new token embeddings** after one pass of self-attention.
- Each row Z_i now carries information not just about $\mathbf{x}_i$ but also about whichever other token(s) it paid attention to.
- In a Transformer model, you'd feed Z into further layers (e.g. feed-forward, residual, multi-head, etc.)—but at its core, self-attention has already given you **context-aware** representations.

Sample - Full Solution

STUDENT NAME:

LAZIZBEK KAHRAMONOV

STUDENT ID NUMBER:

U1710021

SECTION NUMBER IN THIS COURSE:

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Part 3. Calculate One Pass of Self-Attention.

[/3+2]

Token embeddings matrix:

$$X = \begin{bmatrix} 0 & 0 \\ 2 & 1 \end{bmatrix}$$

$$x_1 = [0, 0]$$

$$x_2 = [2, 1]$$

Fixed projection matrices:

$$W^Q = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad W^K = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad W^V = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

Step 1. Compute Q, K, V

$$Q = X W^{Q^T}, \quad K = X W^{K^T}, \quad V = X W^{V^T}.$$

$$Q = \underbrace{\begin{bmatrix} 0 & 0 \\ 2 & 1 \end{bmatrix}}_X \cdot \underbrace{\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}}_{W^{Q^T}} = \begin{bmatrix} 0 & 0 \\ 2 & 1 \end{bmatrix}$$

$$K = \begin{bmatrix} 0 & 0 \\ 2 & 1 \end{bmatrix} \cdot \underbrace{\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}}_{W^{K^T}} = \begin{bmatrix} 0 & 0 \\ 1 & 2 \end{bmatrix}$$

$$V = \begin{bmatrix} 0 & 0 \\ 2 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 3 & 1 \end{bmatrix}$$

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Step 2. Form the Score Matrix S

$$S = Q K^T, \quad S_{ij} = \mathbf{q}_i \cdot \mathbf{k}_j.$$

$$K = \begin{bmatrix} 0 & 0 \\ 1 & 2 \end{bmatrix} \quad K^T = \begin{bmatrix} 0 & 1 \\ 0 & 2 \end{bmatrix}$$

$$S = \begin{bmatrix} 0 & 0 \\ 2 & 1 \end{bmatrix} \cdot \begin{bmatrix} 0 & 1 \\ 0 & 2 \end{bmatrix} = \begin{bmatrix} 0 \cdot 0 + 0 \cdot 0 & 0 \cdot 1 + 0 \cdot 2 \\ 2 \cdot 0 + 1 \cdot 0 & 2 \cdot 1 + 1 \cdot 2 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 4 \end{bmatrix}$$

Step 3. Scale the Scores

$$\tilde{S} = \frac{S}{\sqrt{d_k}} \quad \text{with } d_k = 2,$$

$$\tilde{S} = \frac{S}{\sqrt{2}} = \begin{bmatrix} 0 & 0 \\ 0 & 4 \end{bmatrix} / \sqrt{2} = \begin{bmatrix} 0 & 0 \\ 0 & 2.8284 \end{bmatrix}$$

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Step 4. Convert to Attention Weights

 $A = \text{softmax}(\tilde{S})$ (apply softmax **row-wise**).

$$\text{Row 1)} [0, 0] \rightarrow [e^0, e^0] \rightarrow [1, 1], \text{ sum} = 1 + 1 = 2$$

$$A_1 = \left[\frac{1}{2}, \frac{1}{2} \right] = [0.5, 0.5]$$

$$\text{Row 2)} [0, 2.8284] \rightarrow [e^0, e^{2.8284}] \rightarrow [1, 16.9188]$$

sum ≈ 17.9188

$$A_2 = \left[\frac{1}{17.9188}, \frac{16.9188}{17.9188} \right] \approx [0.0558, 0.9442]$$

Step 5. Compute the Output Z

$$A = \begin{bmatrix} 0.5000 & 0.5000 \\ 0.0558 & 0.9442 \end{bmatrix}$$

$$Z = AV, \quad Z_i = \sum_{j=1}^2 A_{ij} v_j,$$

$$Z = \begin{bmatrix} 0.5 & 0.5 \\ 0.0558 & 0.9442 \end{bmatrix} \cdot \begin{bmatrix} 0 & 0 \\ 3 & 1 \end{bmatrix} = \begin{bmatrix} 0.5 \cdot 0 + 0.5 \cdot 3, & 0.5 \cdot 0 + 0.5 \cdot 1 \\ 0.0558 \cdot 0 + 0.9442 \cdot 3, & 0.0558 \cdot 0 + 0.9442 \cdot 1 \end{bmatrix}$$

$$\approx \begin{bmatrix} 1.5 & 0.5 \\ 2.8326 & 0.9442 \end{bmatrix}$$

Original Embeddings

$$\begin{bmatrix} 0 & 0 \\ 2 & 1 \end{bmatrix}$$

Self-attention
updated embeddings

$$\begin{bmatrix} 1.5 & 0.5 \\ 2.8326 & 0.9442 \end{bmatrix}$$

What to submit

Scan all the pages, make a PDF named <YourStudentID>_MA_assignment2.pdf and submit to eclass.

Bring the hard copy to the first lecture after the deadline or submit it to the Professor's room during office hours.